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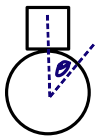
Lecture 18 March 2, 2026

Review of last week:

- Curvilinear 1D systems

  Ex:



cube on a block, find $U(\theta)$ & equilibria

- Central forces: $\vec{F}(\vec{r}) = f(\vec{r})\,\hat{r}$
  - Demonstrated that spherically symmetric central forces are conservative (& vice versa)
  - Introduced spherical coordinates & $d\vec{r}$ in terms of $\hat{r}, \hat{\theta}, \hat{\phi}$
  - found $\vec{\nabla} f$ in spherical coordinates

Activity: Showed $\vec{\nabla}_1 U(\vec{r}_1 - \vec{r}_2) = -\vec{\nabla}_2 U(\vec{r}_1 - \vec{r}_2)$

Showed $\Delta E = \Delta T + \Delta U_{ext}$ for a rigid body

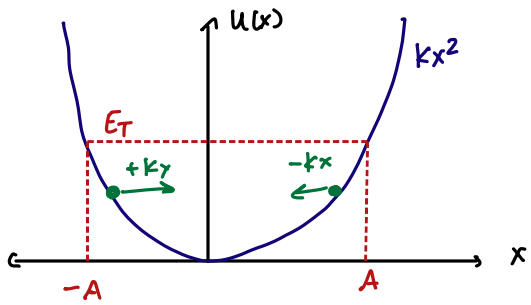
Practiced $\vec{\nabla} \times \vec{F}$, finding $U(\vec{r})$ from $\vec{F}$ using a line integral, checking $\vec{\nabla} U = -\vec{F}$

Chapter 5: Oscillations

A mass on a spring obeys Hook's Law $F_x = -kx$, $k > 0$ & exhibits simple harmonic motion (SHM)

The potential energy is given by:

$$U(x) = \tfrac{1}{2} k x^2$$



SHM is very general

Important property:

- stable equilibrium & restorative force

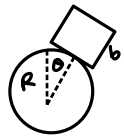
Consider some general function $U(x)$ that has a stable equilibrium at $x=0$:

Near $x=0$:

$$U(x) = \underbrace{U(0)}_{\text{constant, not physical}} + \underbrace{U'(0)\,x}_{\text{at } x=0\text{, there is an equilibrium } \Rightarrow U'(0)=0} + \underbrace{\tfrac{1}{2} U''(0)}_{\text{just some constant (should be positive for the equilibrium to be stable)}}\,x^2 + \dots$$

$$U(x) = \tfrac{1}{2} k x^2 + O(x^3) \quad \text{in general!}$$

Example:



We found $U(\theta) = mg\left[(r+b)\cos\theta + r\theta\sin\theta\right]$ $\qquad \theta = 0$ is an equilibrium

If $\theta \ll 1$, then $\cos\theta \approx 1 - \frac{1}{2}\theta^2$ and $\sin\theta \approx \theta$

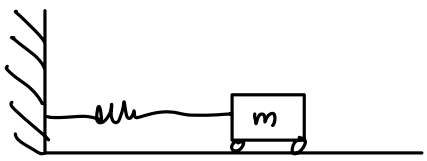
$$\Rightarrow U(\theta) \approx mg(r+b) - \tfrac{1}{2}mg(r+b)\theta^2 + mgr\,\theta^2$$

$$U(\theta) \approx \underbrace{mg(r+b)}_{U_0} + \underbrace{\tfrac{1}{2}mg(r-b)\theta^2}_{\frac{1}{2}k\theta^2}$$

$k = mg(r-b)$

and $k > 0$ when $r > b$, stable equilibrium condition from last week!

Simple Harmonic Motion



$\vec{F} = m\vec{a}$

$-kx = m\ddot{x}$

$-\frac{k}{m}x = \ddot{x}$ $\qquad$ let $\omega = \sqrt{\frac{k}{m}}$ $\Rightarrow$ $\underbrace{\ddot{x} = -\omega^2 x}_{\text{EOM}}$

We must solve this 2nd order dif eq to find $x(t)$:

$\ddot{x} = -\omega^2 x$

- $\exists$ two linearly independent solutions
- Any solution can be written as a linear combo of those two (superposition principle)

Exponential solutions:

$$\left.\begin{aligned} \ddot{x} &= (i\omega)(i\omega)x \Rightarrow x = e^{i\omega t} \\ \ddot{x} &= (-i\omega)(-i\omega)x \Rightarrow x = e^{-i\omega t} \end{aligned}\right\} \text{two independent solutions!}$$

General Solution: $x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$

Sine & cosine solutions:

$$\begin{aligned} \ddot{x} &= (-\omega)(+\omega)x \Rightarrow x = \sin\omega t \\ \ddot{x} &= (+\omega)(-\omega)x \Rightarrow x = \cos\omega t \end{aligned}$$

General solution: $x(t) = B_1\cos\omega t + B_2\sin\omega t$

Using Euler's formula, these are the same: $e^{i\theta} = \cos\theta + i\sin\theta$

$$\Rightarrow x(t) = C_1\left[\cos\omega t + i\sin\omega t\right] + C_2\left[\cos\omega t - i\sin\omega t\right]$$
$$= (C_1 + C_2)\cos\omega t + i(C_1 - C_2)\sin\omega t$$

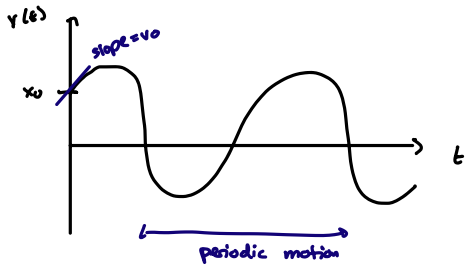
$$\Rightarrow B_1 = C_1 + C_2 \quad \text{and} \quad B_2 = i(C_1 - C_2)$$

We use $x(t) = B_1 \cos\omega t + B_2 \sin\omega t$ to <u>define</u> SHM. $x(t) \in \mathbb{R}$, $\cos\omega t, \sin\omega t \in \mathbb{R}$ $\Rightarrow$ B_1 and $B_2 \in \mathbb{R}$

At $t=0$:

$$x(0) = B_1 + 0 = B_1 \Rightarrow \boxed{B_1 \equiv x_0}$$

$$x'(0) = -\omega B_1 \sin\omega t + \omega B_2 \cos\omega t \Big|_{t=0} = \omega B_2 \equiv v_0 \Rightarrow \boxed{B_2 = \frac{v_0}{\omega}}$$



$$\omega = \frac{2\pi}{\tau} \Rightarrow \tau = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$$

Note that $E = T + U$ = constant

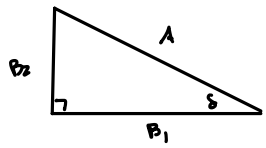
$U = \frac{1}{2}kx^2$ At $t=0 \Rightarrow U = \frac{1}{2}kB_1^2$

$T = \frac{1}{2}m\dot{x}^2$ At $t=0 \Rightarrow T = \frac{1}{2}m\omega^2 B_2^2 = \frac{1}{2}kB_2^2$

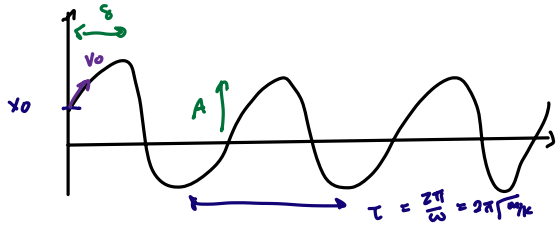
$\Rightarrow E = \frac{1}{2}k(B_1^2 + B_2^2)$ = constant. At turning points $\dot{x} = 0$ and $x = A$

$\Rightarrow E = U + 0 \Rightarrow \frac{1}{2}k(B_1^2 + B_2^2) = \frac{1}{2}kA^2$

$$\Rightarrow \boxed{A^2 = B_1^2 + B_2^2}$$



In general,

$$x(t) = A\left[\frac{B_1}{A}\cos\omega t + \frac{B_2}{A}\sin\omega t\right] = A\underbrace{\left[\cos\delta\cos\omega t + \sin\delta\sin\omega t\right]}_{\text{angle addition formula}} = A\cos(\omega t - \delta)$$

A: amplitude of oscillation; δ: phase shift